

路然雅思

阅读 讲义

目录

第一章 填空题	1
第二章 判断题	9
第三章 匹配题	17
第一节 人名匹配技巧+例题	17
第二节 段落匹配技巧+例题	20
第三节 半句匹配技巧+例题	22
第四章 HEADING 题	25
第五章 选择题	28
第六章 模拟训练	33
第一节 模拟训练一	33
第二节 模拟训练二	40
第三节 模拟训练三	42

阅读

第一章 填空题

一、填空题的“痛点”

(一) 定位不准

帮助定位的 tips:

1. 人名
2. 数字
3. 双引号、斜体字
4. 大写、时间
5. 具象的东西

(二) 同意替换不敏感

1. 整理同义替换
2. 明白同意替换的路径
 - (1) 词到词
 - (2) 词到词组
 - (3) 词到句子
 - (4) 词组到句子
 - (5) 句子到句子

(三) 题干看不懂

在做题之前第一件事情就是看题目！

词汇量！

(四) 空格不知道是词性

大部分都是名词，当然也有动词、数词、形容词等，具体题目具体分析。

二、填空题的分类

(一) 单句填空题

(二) 无选项的段落填空题

(三) 有选项填空题

(四) 简答题

(五) 示意图题目

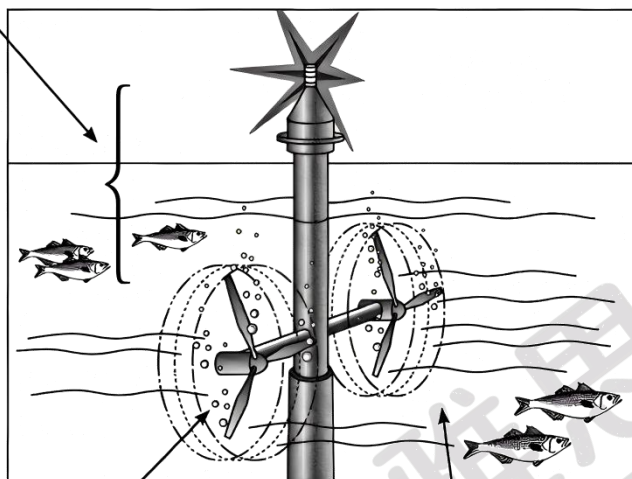
Choose **NO MORE THAN TWO WORDS AND/OR A NUMBER** from the passage for each answer.

Write your answers in boxes 7–10 on your answer sheet.

- 7 For what period of time has hearing loss in schoolchildren been studied in New Zealand?
- 8 In addition to machinery noise, what other type of noise can upset children with autism?
- 9 What term is used to describe the hearing problems of schoolchildren which have not been diagnosed?
- 10 What part of the New Zealand Disability Strategy aims to give schoolchildren equal opportunity?

An Undersea Turbine

Whole tower can be raised
for **23** and the extraction
of seaweed from the blades



Air bubbles result from
the **25** behind blades.
This is known as **26**

Sea life not in danger due to the
fact that blades are comparatively
24

Test 3

Questions 9–12

Complete the summary using the list of words, **A–I**, below.

Write the correct letter, **A–I**, in boxes 9–12 on your answer sheet.

The language debate

According to **9**, there is only one correct form of language. Linguists who take this approach to language place great importance on grammatical **10**
Conversely, the view of **11**, such as Joseph Priestley, is that grammar should be based on **12**

- | | | |
|---------------------------|---------------------------|-------------------------|
| A descriptivists | B language experts | C popular speech |
| D formal language | E evaluation | F rules |
| G modern linguists | H prescriptivists | I change |

Test 4

Questions 7–13

Complete the notes below.

Choose **ONE WORD** from the passage for each answer.

Write your answers in boxes 7–13 on your answer sheet.

Marie Curie's research on radioactivity

- When uranium was discovered to be radioactive, Marie Curie found that the element called **7** had the same property.
- Marie and Pierre Curie's research into the radioactivity of the mineral known as **8** led to the discovery of two new elements.
- In 1911, Marie Curie received recognition for her work on the element **9**
- Marie and Irène Curie developed X-radiography which was used as a medical technique for **10**
- Marie Curie saw the importance of collecting radioactive material both for research and for cases of **11**
- The radioactive material stocked in Paris contributed to the discoveries in the 1930s of the **12** and of what was known as artificial radioactivity.
- During her research, Marie Curie was exposed to radiation and as a result she suffered from **13**

Questions 24–26

Complete the summary below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 24–26 on your answer sheet.

How children acquire a sense of identity

First, children come to realise that they can have an effect on the world around them, for example by handling objects, or causing the image to move when they face a **24** This aspect of self-awareness is difficult to research directly, because of **25** problems.

Secondly, children start to become aware of how they are viewed by others. One important stage in this process is the visual recognition of themselves which usually occurs when they reach the age of two. In Western societies at least, the development of self awareness is often linked to a sense of **26** , and can lead to disputes.

三、填空题怎么做

(一) 定位

用题干的定位词定位文章，因为填空题绝大部分都是顺序题！题文同序。

(二) 找准定位句

找到题干定位词所在位置，并阅读定位句。

(三) 找准同义替换

看懂题干的意思找准同义替换，并根据题干所缺的内容从文章当中找答案。

(四) 根据题目要求来填写句子

如果是一个词那就是一个词，两个词就两个。

READING PASSAGE 1

You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.

The thylacine

The extinct thylacine, also known as the Tasmanian tiger, was a marsupial* that bore a superficial resemblance to a dog. Its most distinguishing feature was the 13-19 dark brown stripes over its back, beginning at the rear of the body and extending onto the tail. The thylacine's average nose-to-tail length for adult males was 162.6 cm, compared to 153.7 cm for females.

The thylacine appeared to occupy most types of terrain except dense rainforest, with open eucalyptus forest thought to be its prime habitat. In terms of feeding, it was exclusively carnivorous, and its stomach was muscular with an ability to distend so that it could eat large amounts of food at one time, probably an adaptation to compensate for long periods when hunting was unsuccessful and food scarce. The thylacine was not a fast runner and probably caught its prey by exhausting it during a long pursuit. During long-distance chases, thylacines were likely to have relied more on scent than any other sense. They emerged to hunt during the evening, night, and early morning and tended to retreat to the hills and forest for shelter during the day. Despite the common name 'tiger', the thylacine had a shy, nervous temperament. Although mainly nocturnal, it was sighted moving during the day and some individuals were even recorded basking in the sun.

The thylacine had an extended breeding season from winter to spring, with indications that

some breeding took place throughout the year. The thylacine, like all marsupials, was tiny and hairless when born. Newborns crawled into the pouch on the belly of their mother, and attached themselves to one of the four teats, remaining there for up to three months. When old enough to leave the pouch, the young stayed in a lair such as a deep rocky cave, well-hidden nest or hollow log, whilst the mother hunted.

Approximately 4,000 years ago, the thylacine was widespread throughout New Guinea and most of mainland Australia, as well as the island of Tasmania. The most recent, well-dated occurrence of a thylacine on the mainland is a carbon-dated fossil from Murray Cave in Western Australia which is around 3,100 years old. Its extinction coincided closely with the arrival of wild dogs called dingoes in Australia and a similar predator in New Guinea. Dingoes never reached Tasmania, and most scientists see this as the main reason for the thylacine survival there.

The dramatic decline of the thylacine in Tasmania, which began in the 1830s and continued for a century, is generally attributed to the relentless efforts of sheep farmers and bounty hunters with shotguns. While this determined campaign undoubtedly played a large part, it is likely that various other factors also contributed to the decline and eventual extinction of the species. These include competition with wild dogs introduced by European settlers, loss of habitat along with the disappearance of prey species, and a distemper-like disease which may also have affected the thylacine.

Questions 1–5

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 1–5 on your answer sheet.

The thylacine

Appearance and behaviour

- looked rather like a dog
- had a series of stripes along its body and tail
- ate an entirely 1 diet
- probably depended mainly on 2 when hunting
- young spent first months of life inside its mother's 3

Decline and extinction

- last evidence in mainland Australia is a 3,100-year-old 4
- probably went extinct in mainland Australia due to animals known as dingoes
- reduction in 5 and available sources of food were partly responsible for decline in Tasmania

四、技巧总结

(一) 定位

(二) 找准定位词，熟悉同义替换

(三) 根据缺少的成分来挑选答案

Test 4

Rocha and his team used state-of-the-art ultrasonic recorders to record over a thousand bat 'feeding buzzes' (echolocation sequences used by bats to target their prey) at 54 sites, in order to identify the favourite feeding spots of the bats. They next used DNA barcoding techniques to analyse droppings collected from bats at the different sites.

The recordings revealed that bat activity over rice fields was much higher than it was in continuous forest—seven times higher over rice fields which were on flat ground, and sixteen times higher over fields on the sides of hills—leaving no doubt that the animals are preferentially foraging in these man-made ecosystems. The researchers suggest that the bats favour these fields because lack of water and nutrient run-off make these crops more susceptible to insect pest infestations. DNA analysis showed that all six species of bat had fed on economically important insect pests. While the findings indicated that rice farming benefits most from the bats, the scientists also found indications that the bats were consuming pests of other crops, including the black twig borer (which infests coffee plants), the sugarcane cicada, the macadamia nut-borer, and the sober tabby (a pest of citrus fruits).

'The effectiveness of bats as pest controllers has already been proven in the USA and Catalonia,' said co-author James Kemp, from the University of Lisbon. 'But our study is the first to show this happening in Madagascar, where the stakes for both farmers and conservationists are so high.'

Local people may have a further reason to be grateful to their bats. While the animals often associated with spreading disease, Rocha and his team found evidence that Malagasy bats feed not just on crop pests but also on mosquitoes—carriers of malaria, Rift Valley fever virus and elephantiasis—as well as black flies, which spread river blindness.

Rocha points out that the relationship is complicated. When food is scarce, bats become a crucial source of protein for local people. Even the children will hunt them. And as well as roosting in trees, the bats sometimes roost in buildings, but are not welcomed there because they make them unclean. At the same time, however, they are associated with sacred caves and the ancestors, so they can be viewed as beings between worlds, which makes them very significant in the culture of the people. And one potential problem is that while these bats are benefiting from farming, at the same time deforestation is reducing the places where they can roost, which could have long-term effects on their numbers. Rocha says, 'With the right help, we hope that farmers can promote this mutually beneficial relationship by installing bat houses.'

Rocha and his colleagues believe that maximising bat populations can help to boost crop yields and promote sustainable livelihoods. The team is now calling for further research to quantify this contribution. 'I'm very optimistic,' says Rocha. 'If we give nature a hand, we can speed up the

process of regeneration.

Questions 7–13

Complete the table below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 7–13 on your answer sheet.

The study carried out by Rocha's team	
Aim	to investigate the feeding habits of bats in farmland near the Ranomafana National Park
Method	- ultrasonic recording to identify favourite feeding spots - DNA analysis of bat 7
Findings	- the bats - were most active in rice fields located on hills - ate pests of rice, 8, sugarcane, nuts and fruit - prevent the spread of disease by eating 9 and blackflies - local attitudes to bats are mixed: - they provide food rich in 10 - the buildings where they roost become 11 - they play an important role in local 12
Recommendation	farmers should provide special 13 to support the bat population

第二章 判断题

判断题

YES/NO/NG

T/F/NG

一、判断题“痛点”

(一) 不知道定位在哪里

帮助定位的 tips:

- 1.人名
- 2.数字
- 3.双引号、斜体字
- 4.大写、时间
- 5.具象的东西

(二) 分不清楚 F/NO 和 NG

弄清楚错误和没给之间的关系。

YES/TRUE 的话那就是同义替换之间的关系,以及正话反说或者双重否定表示肯定的句子。

(三) 顺序题但是少量乱序

与填空题类似,都是以顺序题目的眼光去做判断题。

二、判断题的特征

(一) YES/TRUE

对同一句话的改写:

比如说“我依靠你”=“我依赖你”。

同句子的总结或者归纳：

比如说文章中说“我不能没有水！”，题目说“水对我很重要”。

(二) NO/FALSE

关系不一致或者是反义词。

关系不一致：

原文：我的苹果比你的要多。

题目：我跟你的苹果一样多。

反义词：

原文：中国正在发展。

题目：中国实力正在减弱。

原文

A It is difficult to conceive of vigorous economic growth without an efficient transport system.[Although modern information technologies can reduce the demand for physical transport by facilitating teleworking and teleservices, the requirement for transport continues to increase.]There are two key factors behind this trend.For passenger

题目

22 The need for transport is growing,despite technological developments.

原文

Without a qualification from a reputable school or university, it is unlikely to find a good job.

题目

It is impossible to get a good job without a qualification from a respected institution.

(三) NOT GIVEN

单词找不到或者关系找不到

为什么会出现 NOT GIVEN?

主要是考官在题目当中添加的信息:

1. 单词型

2. 关系型

举个小例子,

原文: 想考 A 的方法是 B

题目: 想考 A 的方法是 C

原文: A 在比赛之中获得第一名

题目: A 在比赛当中获得第二名

原文: A 参加了比赛

题目: A 在比赛之中表现的比 B 好

原文: 做成 A 的最好方法是 B

题目: 通过 C 我们可以做成 A

小 TIPS: 每次遇到 F/NG 的时候问问自己题目和文章是否矛盾, 不矛盾的话, 就是 NG

矛盾的话就是 F

QUESTIONS:

1. In the future, governments should maintain ownership of water infrastructures.

(原文当中从未出现过 government 这一个单词)

2. MIRTHP hoped to improve the movement of goods from the Maket district to the country's capital.

(但是全文当中并没有出现 CAPITAL 一词)

小试牛刀:

Q: Private schools in Japan are more modern and spacious than state-run lower secondary schools.

PASSAGE:

Lower secondary schools in Japan cover three school years, from the seventh grade (age13) to the ninth grade (age15). Virtually all pupils at this stage attend state schools: only 3 percent are in the private sector, schools are usually modern in design, set well back from the road and spacious inside.

(四) 总结

一般来说 NG 是具体信息找不到或者是关系找不到。

Such age-defying feats are rare in the animal kingdom. Naked mole rats can live for almost 30 years and stay fit for nearly their entire lives. They can still reproduce even when old, and they never get cancer. But the vast majority of animals deteriorate with age just like people do. Like the naked mole rat, ants are social creatures that usually live in highly organised colonies. 'It's this social complexity that makes *P. dentata* useful for studying aging in people,' says Giraldo, now at the California Institute of Technology. Humans are also highly social, a trait that has been connected to healthier aging. By contrast, most animal studies of aging use mice, worms or fruit flies, which all lead much more isolated lives.

Questions 9–13

Do the following statements agree with the information given in Reading Passage 1?

In boxes 9–13 on your answer sheet, write

TRUE	if the statement agrees with the information
FALSE	if the statement contradicts the information
NOT GIVEN	if there is no information on this

- 9 *Pheidole dentata* ants are the only known animals which remain active for almost their whole lives.
- 10 Ysabel Giraldo was the first person to study *Pheidole dentata* ants using precise data about the insects' ages.
- 11 The ants in Giraldo's experiments behaved as she had predicted that they would.
- 12 The recent studies of bees used different methods of measuring age-related decline.
- 13 *Pheidole dentata* ants kept in laboratory conditions tend to live longer lives.

B Short is calling for a sweeping reinvention of how skyscrapers and major public buildings are designed -to end the reliance on sealed buildings which exist solely via the 'life support' system of vast air conditioning units.

Instead, he shows it is entirely possible to accommodate natural ventilation and cooling in large buildings by looking into the past, before the widespread introduction of air conditioning

systems, which were 'relentlessly and aggressively marketed' by their inventors.

4. Slow language development in children is likely to prove disappointing to their parents.

- YES
- NO
- NOT GIVEN

Henderson learned photography in Montreal around the year 1857 and quickly took it up as a serious amateur. He became a personal friend and colleague of the Scottish-Canadian photographer William Notman. The two men made a photographic excursion to Niagara Falls in 1860 and they cooperated on experiments with magnesium flares as a source of artificial light in 1865. They belonged to the same societies and were among the founding members of the Art Association of Montreal. Henderson acted as chairman of the association's first meeting, which was held in Notman's studio on 11 January 1860.

3. Henderson and Notman were surprised by the results of their 1865 experiment.

- TRUE
- FALSE
- NOT GIVEN

In 1866, he gave up his business to open a photographic studio, advertising himself as a portrait and landscape photographer. From about 1870 he dropped portraiture to specialize in landscape photography and other views. His numerous photographs of city life revealed in street scenes, houses, and markets are alive with human activity; and although his favourite subject was landscape he usually composed his scenes around such human pursuits as farming the land, cutting ice on a river, or sailing down a woodland stream. There was sufficient demand for these types of scenes and others he took depicting the lumber trade, steamboats and waterfalls to enable him to make a living. There was little competing hobby or amateur photography before the late 1880s because of the time-consuming techniques involved and the weight of the equipment.

People wanted to buy photographs as souvenirs of a trip or as gifts, and catering to this market, Henderson had stock photographs on display at his studio for mounting, framing, or inclusion in albums.

5. The studio that Henderson opened in 1866 was close to his home.

- TRUE
- FALSE
- NOT GIVEN

Alexander Henderson was born in Scotland in 1831 and was the son of a successful merchant. His grandfather, also called Alexander, had founded the family business, and later became the first chairman of the National Bank of Scotland. The family had extensive landholdings in Scotland. Besides its residence in Edinburgh, it owned Press Estate, 650 acres of farmland about 35 miles southeast of the city. The family often stayed at Press Castle, the large mansion on the northern edge of the property, and Alexander spent much of his childhood in the area, playing on the beach near Eyemouth or fishing in the streams nearby.

1. Henderson rarely visited the area around Press estate when he was younger.

- TRUE
- FALSE
- NOT GIVEN

In spite of their friendship, their styles of photography were quite different. While Notman's landscapes were noted for their bold realism, Henderson for the first 20 years of his career produced romantic images, showing the strong influence of the British landscape tradition. His artistic and technical progress was rapid and in 1865 he published his first major collection of landscape photographs. The publication had limited circulation (only seven copies have ever been found), and was called Canadian Views and Studies. The contents of each copy vary significantly and have proved a useful source for evaluating Henderson's early work.

4. There were many similarities between Henderson's early landscapes and those of Notman.

- TRUE
- FALSE
- NOT GIVEN

In the 1870s and 1880s Henderson travelled widely throughout Quebec and Ontario, in Canada, documenting the major cities of the two provinces and many of the villages in Quebec. He was especially fond of the wilderness and often travelled by canoe on the Blanche, du Lièvre, and other noted eastern rivers. He went on several occasions to the Maritimes and in 1872 he sailed by yacht along the lower north shore of the St Lawrence River. That same year, while in the lower St Lawrence River region, he took some photographs of the construction of the Intercolonial Railway. This undertaking led in 1875 to a commission from the railway to record the principal structures along the almost-completed line connecting Montreal to Halifax. Commissions from other railways followed. In 1876 he photographed bridges on the Quebec, Montreal, Ottawa and Occidental Railway between Montreal and Ottawa. In 1885 he went west along the Canadian Pacific Railway (CPR) as far as Rogers Pass in British Columbia, where he took photographs of the mountains and the progress of construction.

7. When Henderson began work for the Intercolonial Railway, the Montreal to Halifax line had been finished.

- TRUE
- FALSE
- NOT GIVEN

Even after he went to school at Murcheston Academy on the outskirts of Edinburgh, Henderson returned to Press at weekends. In 1849 he began a three-year apprenticeship to become an accountant. Although he never liked the prospect of a business career, he stayed with it to please his family. In October 1855, however, he emigrated to Canada with his wife Agnes Elder Roberts and they settled in Montreal.

2. Henderson pursued a business career because it was what his family wanted.

- TRUE
- FALSE
- NOT GIVEN

In 1892 Henderson accepted a fulltime position with the CPR as manager of a photographic department which he was to set up and administer. His duties included spending four months in the field each year. That summer he made his second trip west, photographing extensively along the

railway line as far as Victoria. He continued in this post until 1897, when he retired completely from photography.

8. Henderson's last work as a photographer was with the Canadian Pacific Railway.

- TRUE
- FALSE
- NOT GIVEN

三、技巧总结

- (一) 对的在关系一致的前提之下一定有转化或者改写
- (二) 错的要不就是反义词，要不就是关系不对
- (三) 没给的两组关系，单词和关系



第三章 匹配题

第一节 人名匹配技巧+例题

一、人名匹配题

- (一) 圈出所有人名
- (二) 划出题干的关键词
- (三) 题干和文章进行比对
- (四) 找到近义词或者句意转换
- (五) 消极和积极的单词划上上下下箭头

二、做题难点

- (一) 定位好找但是答案不好找?
- (二) 难道只需要看定位句吗?
- (三) 顺序还是乱序?

NB You may use any letter more than once.

- 7 It is desirable for the same possibilities to be open to everyone.
- 8 No section of society should have preferential treatment at the expense of another.
- 9 People should only gain benefits on the basis of what they actually achieve.
- 10 Variation in intelligence begins at birth.
- 11 The more intelligent people should be in positions of power.
- 12 Everyone can develop the same abilities.
- 13 People of low intelligence are likely to lead uncontrolled lives.

List of Theories

- A Hamiltonian
- B Jeffersonian
- C Jacksonian

F I have suggested that there are three major implicit theories of how intelligence relates to society as a whole (Sternberg.1997). These might be called Hamiltonian, Jeffersonian, and Jacksonian. These views are not based strictly, but rather, loosely, on the philosophies of Alexander

Hamilton, Thomas Jefferson, and Andrew Jackson, three great statesmen in the history of the United States.

G The Hamiltonian view, which is similar to the Platonic view, is that people are born with different levels of intelligence and that those who are less intelligent need the good offices of the more intelligent to keep them in line, whether they are called government officials or, in Plato's term, philosopher-kings. Herrnstein and Murray (1994) seem to have shared this belief when they wrote about the emergence of a cognitive (high-IQ) elite, which eventually would have to take responsibility for the largely irresponsible masses of non-elite (low-IQ) people who cannot take care of themselves. Left to themselves, the unintelligent would create, as they always have created, a kind of chaos.

H The Jeffersonian view is that people should have equal opportunities, but they do not necessarily avail themselves equally of these opportunities and are not necessarily equally rewarded for their accomplishments. People are rewarded for what they accomplish, if given equal opportunity. Low achievers are not rewarded to the same extent as high achievers. In the Jeffersonian view, the goal of education is not to favor or foster an elite, as in the Hamiltonian tradition, but rather to allow children the opportunities to make full use of the skills they have. My own views are similar to these (Sternberg, 1997).

I The Jacksonian View is that all people are equal, not only as human beings but in terms of their competencies - that one person would serve as well as another in government or on a jury or in almost any position of responsibility. In this view of democracy, people are essentially intersubstitutable except for specialized skills, all of which can be learned. In this view, we do not need or want any institutions that might lead to favoring one group over another.

Questions 27–31

Look at the following statements (Questions 27–31) and the list of researchers below.

Match each statement with the correct researcher, **A–G**.

Write the correct letter, **A–G**, in boxes 27–31 on your answer sheet.

- 27 Play can be divided into a number of separate categories.
- 28 Adults' intended goals affect how they play with children.
- 29 Combining work with play may be the best way for children to learn.
- 30 Certain elements of play are more significant than others.
- 31 Activities can be classified on a scale of playfulness.

List of Researchers

- A** Elkind
- B** Miller & Almon
- C** Rubin et al.
- D** Stuart Brown
- E** Pellegrini
- F** Joan Goodman
- G** Hirsch-Pasek et al.

Yet, while experts continue to expound a powerful argument for the importance of play in

children's lives, the actual time children spend playing continues to decrease. Today, children play eight hours less each week than their counterparts did two decades ago (Elkind 2008). Under pressure of rising academic standards, play is being replaced by test preparation in kindergartens and grade schools, and parents who aim to give their preschoolers a leg up are led to believe that flashcards and educational 'toys' are the path to success. Our society has created a false dichotomy between play and learning.

Through play, children learn to regulate their behavior, lay the foundations for later learning in science and mathematics, figure out the complex negotiations of social relationships, build a repertoire of creative problem-solving skills, and so much more. There is also an important role for adults in guiding children through playful learning opportunities.

Full consensus on a formal definition of play continues to elude the researchers and theorists who study it. Definitions range from discrete descriptions of various types of play such as physical, construction, language, or symbolic play (Miller & Almon 2009), to lists of broad criteria, based on observations and attitudes, that are meant to capture the essence of all play behaviors (e.g. Rubin et al. 1983).

A majority of the contemporary definitions of play focus on several key criteria. The founder of the National Institute for Play, Stuart Brown, has described play as 'anything that spontaneously is done for its own sake'. More specifically, he says it 'appears purposeless, produces pleasure and joy, [and] leads one to the next stage of mastery' (as quoted in Tippett 2008). Similarly, Miller and Almon (2009) say that play includes 'activities that are freely chosen and directed by children and arise from intrinsic motivation'. Often, play is defined along a continuum as more or less playful using the following set of behavioral and dispositional criteria (e.g. Rubin et al. 1983).

Play is pleasurable: Children must enjoy the activity or it is not play. It is intrinsically motivated: Children engage in play simply for the satisfaction the behavior itself brings. It has no extrinsically motivated function or goal. Play is process oriented: When children

According to this view, children's playful behaviors can range in degree from 0% to 100% playful. Rubin and colleagues did not assign greater weight to any one dimension in determining playfulness; however, other researchers have suggested that process orientation and a lack of obvious functional purpose may be the most important aspects of play (e.g. Pellegrini 2009).

From the perspective of a continuum, play can thus blend with other motives and attitudes that are less playful, such as work. Unlike play, work is typically not viewed as enjoyable and it is extrinsically motivated (i.e. it is goal oriented). Researcher Joan Goodman (1994) suggested that hybrid forms of work and play are not a detriment to learning; rather, they can provide optimal contexts for learning. For example, a child may be engaged in a difficult, goal-directed activity set up by their teacher, but they may still be actively engaged and intrinsically motivated. At this mid-point between play and work, the child's motivation, coupled with guidance from an adult, can create robust opportunities for playful learning.

Critically, recent research supports the idea that adults can facilitate children's learning while maintaining a playful approach in interactions known as 'guided play' (Fisher et al. 2011). The adult's role in play varies as a function of their educational goals and the child's developmental level (Hirsch-Pasek et al. 2009).

三、技巧总结

- (一) 划出人名，找准定位
- (二) 读懂题目去匹配
- (三) NB?无 NB?
- (四) 正负向词语打上标记

第二节 段落匹配技巧+例题

一、段落匹配

(一) 做题方法

- 1. 类似人名匹配但是没有选项
- 2. 做题方法类似
- 3. 读懂句意
- 4. 放弃废话词
- 5. 乱序题目

Write the correct letter, A-H, in boxes 1-4 on your answer sheet.

- 1 an explanation of the industrial processes that create potential raw materials for concrete
- 2 a reference to the various locations where high-rise wooden buildings can be found
- 3 an indication of how widely available the raw materials of concrete are
- 4 the belief that more high-rise wooden buildings are needed before wood can be regarded as a viable construction material

Not only are the ingredients of concrete relatively cheap and found in abundance in most places around the globe, the stuff itself has marvelous properties: Portland cement, the vital component of concrete, is mouldable and pourable, but quickly sets hard. Cheeseman also notes another advantage: concrete and steel have similar thermal expansion properties, so steel can be used to reinforce concrete, making it far stronger and more flexible as a building material than it could be on its own. According to Cheeseman, all these factors together make concrete hard to beat. 'Concrete is amazing stuff. Making anything with similar properties is going to be very difficult.'

Stora Enso is Europe's biggest supplier of cross-laminated timber, and its vice-president Markus Mannstrom reports that the company is seeing increasing demand globally for building in wood, with climate change concerns the key driver. Finland, with its large forests, where Stora

Enso is based, has been leading the way, but the company is seeing a rise in demand for its timber products across the world, including in Asia. Of course, using timber in a building also locks away the carbon that it absorbed as it grew. But even treated wood has its limitations and only when a wider range of construction projects has been proven in practice will it be possible to see wood as a real alternative to concrete in constructing tall buildings.

A possible alternative to concrete is wood. Making buildings from wood may seem like a rather medieval idea, but climate change is driving architects to turn to treated timber as a possible resource. Recent years have seen the emergence of tall buildings constructed almost entirely from timber. Vancouver, Vienna and Brumunddal in Norway are all home to constructed tall, wooden buildings.

Fly ash and slag from iron ore are possible alternatives to cement in a concrete mix. Fly ash, a byproduct of coal-burning power plants, can be incorporated into concrete mixes to make up as much as 15 to 30% of the cement, without harming the strength or durability of the resulting mix. Iron-ore slag, a byproduct of the iron-ore smelting process, can be used in a similar way. Their incorporation into concrete mixes has the potential to reduce greenhouse gas emissions.

二、技巧总结

(一) 读懂句意

(二) 永远掌握近义词在应试考试的作用

(三) 先易后难

A Two things distinguish food production from all other productive activities: first, every single person needs food each day and has a right to it; and second, it is hugely dependent on nature. These two unique aspects, one political, the other natural, make food production highly vulnerable and different from any other business. At the same time, cultural values are highly entrenched in food and agricultural systems worldwide.

B Farmers everywhere face major risks, including extreme weather long-term climate change, and price volatility in input and product markets. However, smallholder farmers in developing countries must in addition deal with adverse environments, both natural, in terms of soil quality rainfall etc, and human, in terms of infrastructure, financial systems, markets, knowledge and technology. Counter-intuitively, hunger is prevalent among many smallholder farmers in the developing world.

H Some participating authors and commentators argued in favour of community-based and autonomous risk management strategies through collective action groups, co-operatives or producers' groups. Such groups enhance market opportunities for small-scale producers, reduce marketing costs and synchronise buying and selling with seasonal price conditions. According to Murphy, 'collective action offers an important way for farmers to strengthen their political and economic bargaining power, and to reduce their business risks.' One commentator, Giel Ton, warned that collective action does not come as a free good. It takes time, effort and money to organise, build trust and to experiment. Others, like Marcel Vernooij and Marcel Beukeboom, suggested that

in order to' apply what we already know',all stakeholders,including business,government,scientists and civil society,must work together,starting at the beginning of the value chain.

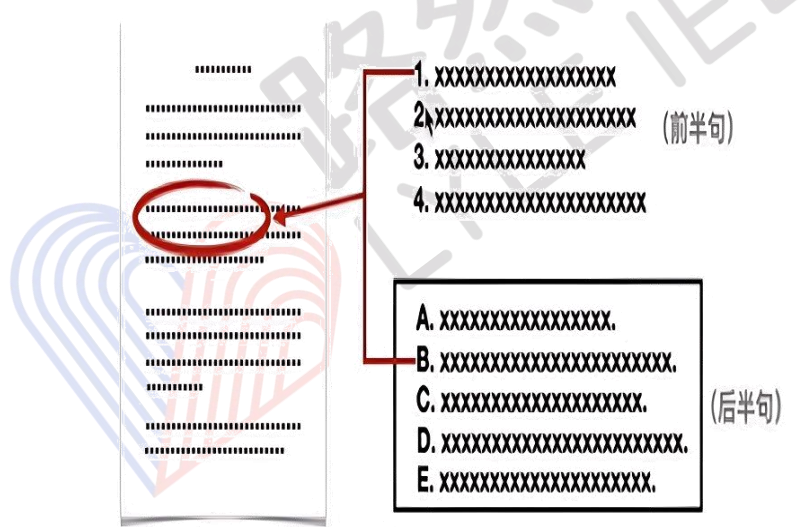
Write the correct letter, A–I, in boxes 1–3 on your answer sheet.

- 1 a reference to characteristics that only apply to food production
- 2 a reference to challenges faced only by farmers in certain parts of the world
- 3 a reference to difficulties in bringing about co-operation between farmers

第三节 半句匹配技巧+例题

一、半句匹配

- (一) 难度较难
- (二) 顺序提
- (三) 先易后难



- 1.正确匹配完整句子，大多是原文几句话的同义替换，所以，不要进行多点定位。
- 2.同义替换能力是关键。
- 3.句子匹配题也是大多数题文同序，不存在 NB 复选情况。

32 For a long time biologists rejected_____

33 Opposing views on evolutionary throwbacks are represented by_____

- 34 Examples of evolutionary throwbacks have led to _____
- 35 The shark and killer whale are mentioned to exemplify _____
- 36 One explanation for the findings of Wagner's research is _____

- A.the question of how certain long-lost trait could reappear.
- B.the occurrence of a particular feature indifferent species.
- C.parallels drawn between behaviour and appearance.
- D.the continued existence of certain genetic information.
- E.the doubts felt about evolutionary throwbacks.
- F.the possibility of evolution being reversible.
- G.Dollo's findings and the convictions held by Lombroso.

The description of any animal as an 'evolutionary throwback' is controversial. For the better part of a century, most biologists have been reluctant to use those words, mindful of a principle of evolution that says 'evolution cannot run backwards'. But as more and more examples come to light and modern genetics enters the scene, that principle is having to be rewritten. Not only are evolutionary throwbacks possible, they sometimes play an important role in the forward march of evolution.

The technical term for an evolutionary throwback is an 'atavism', from the Latin atavus, meaning forefather. The word has ugly connotations thanks largely to Cesare Lombroso, a 19th-century Italian medic who argued that criminals were born not made and could be identified by certain physical features that were throwbacks to a primitive, sub-human state.

While Lombroso was measuring criminals, a Belgian palaeontologist called Louis Dollo was studying fossil records and coming to the opposite conclusion. In 1890 he proposed that evolution was irreversible: that 'an organism is unable to return, even partially, to a previous stage already realised in the ranks of its ancestors'. Early 20th-century biologists came to a similar conclusion, though they qualified it in terms of probability, stating that there is no reason why evolution cannot run backwards—it is just very unlikely. And so the idea of irreversibility in evolution stuck and came to be known as 'Dollo's law'.

If Dollo's law is right, atavisms should occur only very rarely, if at all. Yet almost since the idea took root, exceptions have been cropping up. In 1919, for example, a humpback whale with a pair of leg-like appendages over a metre long, complete with a full set of limb bones, was caught off Vancouver Island in Canada. Explorer Roy Chapman Andrews argued at the time that the whale must be a throwback to a land-living ancestor. 'I can see no other explanation,' he wrote in 1921.

Since then, so many other examples have been discovered that it no longer makes sense to say that evolution is as good as irreversible. And this poses a puzzle: how can characteristics that disappeared millions of years ago suddenly reappear? In 1994, Rudolf Raff and colleagues at Indiana University in the USA decided to use genetics to put a number on the probability of evolution going into reverse. They reasoned that while some evolutionary changes involve the loss of genes and are therefore irreversible, others may be the result of genes being switched off. If these silent genes are somehow switched back on, they argued, long-lost traits could reappear.

Raff's team went on to calculate the likelihood of it happening. Silent genes accumulate random mutations, they reasoned, eventually rendering them useless. So how long can a gene

survive in a species if it is no longer used? The team calculated that there is a good chance of silent genes surviving for up to 6 million years in at least a few individuals in a population, and that some might survive as long as 10 million years. In other words, throwbacks are possible, but only to the relatively recent evolutionary past.

As a possible example, the team pointed to the mole salamanders of Mexico and California. Like most amphibians these begin life in a juvenile ‘tadpole’ state, then metamorphose into the adult form—except for one species, the axolotl, which famously lives its entire life as a juvenile. The simplest explanation for this is that the axolotl lineage alone lost the ability to metamorphose, while others retained it. From a detailed analysis of the salamanders’ family tree, however, it is clear that the other lineages evolved from an ancestor that itself had lost the ability to metamorphose. In other words, metamorphosis in mole salamanders is an atavism. The salamander example fits with Raff’s 10-million-year time frame.

More recently, however, examples have been reported that break the time limit, suggesting that silent genes may not be the whole story. In a paper published last year, biologist Gunter Wagner of Yale University reported some work on the evolutionary history of a group of South American lizards called *Bachia*. Many of these have minuscule limbs; some look more like snakes than lizards and a few have completely lost the toes on their hind limbs. Other species, however, sport up to four toes on their hind legs. The simplest explanation is that the toed lineages never lost their toes, but Wagner begs to differ. According to his analysis of the *Bachia* family tree, the toed species re-evolved toes from toeless ancestors and, what is more, digit loss and gain has occurred on more than one occasion over tens of millions of years.

So what’s going on? One possibility is that these traits are lost and then simply reappear, in much the same way that similar structures can independently arise in unrelated species, such as the dorsal fins of sharks and killer whales. Another more intriguing possibility is that the genetic information needed to make toes somehow survived for tens or perhaps hundreds of millions of years in the lizards and was reactivated. These atavistic traits provided an advantage and spread through the population, effectively reversing evolution.

But if silent genes degrade within 6 to 10 million years, how can long-lost traits be reactivated over longer timescales? The answer may lie in the womb. Early embryos of many species develop ancestral features. Snake embryos, for example, sprout hind limb buds. Later in development, these features disappear thanks to developmental programs that say ‘lose the leg’. If for any reason this does not happen, the ancestral feature may not disappear, leading to an atavism.

二、技巧总结

(一) 顺序题目

(二) 题目和选项都需要阅读

(三) 近义词还是王道

第四章 HEADING 题

一、HEADING

- (一) 考察提炼、归纳的能力
- (二) 考察综合英语实力
- (三) 乱序

二、HEADING 题目的做题要点

- (一) 一如既往，划取关键信息
- (二) 匹配句式转换或者近义词
- (三) 选项和文章进行比对（连连看）

Reading Passage 2 has seven paragraphs, **A–G**.

Choose the correct heading for each paragraph from the list of headings below.

Write the correct number, **i–viii**, in boxes 14–20 on your answer sheet.

List of Headings

- i** Different accounts of the same journey
- ii** Bingham gains support
- iii** A common belief
- iv** The aim of the trip
- v** A dramatic description
- vi** A new route
- vii** Bingham publishes his theory
- viii** Bingham's lack of enthusiasm

- 14** Paragraph **A**
- 15** Paragraph **B**
- 16** Paragraph **C**
- 17** Paragraph **D**
- 18** Paragraph **E**
- 19** Paragraph **F**
- 20** Paragraph **G**

A When the US explorer and academic Hiram Bingham arrived in South America in 1911, he was ready for what was to be the greatest achievement of his life: the exploration of the remote hinterland to the west of Cusco, the old capital of the Inca empire in the Andes mountains of Peru. His goal was to locate the remains of a city called Vitcos, the last capital of the Inca civilisation. Cusco lies on a high plateau at an elevation of more than 3,000 metres, and Bingham's plan was to descend from this plateau along the valley of the Urubamba river, which takes a circuitous route down to the Amazon and passes through an area of dramatic canyons and mountain ranges.

B When Bingham and his team set off down the Urubamba in late July, they had an advantage over travellers who had preceded them: a track had recently been blasted down the valley canyon to enable rubber to be brought up by mules from the jungle. Almost all previous travellers had left the river at Ollantaytambo and taken a high pass across the mountains to rejoin the river lower down, thereby cutting a substantial corner, but also therefore never passing through the area around Machu Picchu.

C On 24 July they were a few days into their descent of the valley. The day began slowly, with Bingham trying to arrange sufficient mules for the next stage of the trek. His companions showed no interest in accompanying him up the nearby hill to see some ruins that a local farmer, Melchor Arteaga, had told them about the night before. The morning was dull and damp, and Bingham also seems to have been less than keen on the prospect of climbing the hill. In his book *Lost City of the Incas*, he relates that he made the ascent without having the least expectation that he would find anything at the top.

D Bingham writes about the approach in vivid style in his book. First, as he climbs up the hill, he describes the ever-present possibility of deadly snakes, 'capable of making considerable springs when in pursuit of their prey'; not that he sees any. Then there's a sense of mounting discovery as he comes across great sweeps of terraces, then a mausoleum, followed by monumental staircases and, finally the grand ceremonial buildings of Machu Picchu. 'it seemed like an unbelievable dream ... the sight held me spellbound ... he wrote.

E We should remember, however, that *Lost City of the Incas* is a work of hindsight not written until 1948, many years after his journey. His journal entries of the time reveal a much more gradual appreciation of his achievement. He spent the afternoon at the ruins noting down the dimensions of some of the buildings then descended and rejoined his companions, to whom he seems to have said little about his discovery. At this stage, Bingham didn't realise the extent or the importance of the site, nor did he realise what use he could make of the discovery.

F However, soon after returning it occurred to him that he could make a name for himself from this discovery. When he came to write the *National Geographic* magazine article that broke the story to the world in April 1913, he knew he had to produce a big idea. He wondered whether it could have been the birthplace of the very first Inca, Manco the Great, and whether it could also have been what chroniclers described as 'the last city of the Incas'. This term refers to Vilcabamba the settlement where the Incas had fled from Spanish invaders in the 1530s. Bingham made desperate attempts to prove this belief for nearly 40 years. Sadly, his vision of the site as both the beginning and end of the Inca civilisation, while a magnificent one, is inaccurate. We now know that Vilcabamba actually lies 65 kilometres away in the depths of the jungle.

G One question that has perplexed visitors, historians and archaeologists alike ever since Bingham, is why the site seems to have been abandoned before the Spanish Conquest. There are no references to it by any of the Spanish chroniclers - and if they had known of its existence so

close to Cusco they would certainly have come in search of gold. An idea which has gained wide acceptance over the past few years is that Machu Picchu was a *moya*, a country estate built by an Inca emperor to escape the cold winters of Cusco, where the elite could enjoy monumental architecture and spectacular views. Furthermore, the particular architecture of Machu Picchu suggests that it was constructed at the time of the greatest of all the Incas, the emperor Pachacuti (c.1438-71). By custom, Pachacuti's descendants built other similar estates for their own use, and so Machu Picchu would have been abandoned after his death, some 50 years before the Spanish Conquest.

三、技巧总结

(一) 读懂句意，划出关键词

(二) 近义词和同意转化

(三) 先易后难



第五章 选择题

一、选择题之单选与多选题

- (一) 题目难度大 (先易后难)
- (二) 顺序题+定位不需要太过在意
- (三) 近义词还是关键 (切记不能主观想象永远遵守文章大意)

二、单选题的做题步骤

- (一) 读懂句意
- (二) 先易后难
- (三) 做题如同判断题
- (四) 注意句式转换+同意替换
- (五) 顺序

Questions 14–18

Choose the correct letter, **A**, **B**, **C** or **D**.

Write the correct letter in boxes 14–18 on your answer sheet.

- 14** In the first paragraph, the writer makes a point that
- A** the director should plan the sound track at an early stage in filming.
 - B** it would be wrong to overlook the contribution of sound to the artistry of films.
 - C** the music industry can have a beneficial influence on sound in film.
 - D** it is important for those working on the sound in a film to have sole responsibility for it.
- 15** One reason that the writer refers to Humphrey Bogart is to exemplify
- A** the importance of the actor and the character appearing to have similar personalities.
 - B** the audience's wish that actors are visually appropriate for their roles.
 - C** the value of the actor having had similar feelings to the character.
 - D** the audience's preference for dialogue to be as authentic as possible.
-

Though we might think of film as an essentially visual experience, we really cannot afford to underestimate the importance of film sound. A meaningful sound track is often as complicated as the image on the screen, and is ultimately just as much the responsibility of the director. The entire

sound track consists of three essential ingredients: the human voice, sound effects and music. These three tracks must be mixed and balanced so as to produce the necessary emphases which in turn create desired effects.

Topics which essentially refer to the three previously mentioned tracks are discussed below. They include dialogue synchronous and asynchronous sound effects, and music.

Let us start with dialogue. As is the case with stage drama, dialogue serves to tell the story and expresses feelings and motivations of characters as well. Often with film characterization the audience perceives little or no difference between the character and the actor. Thus, for example, the actor Humphrey Bogart is the character Sam Spade; film personality and life personality seem to merge.

Perhaps this is because the very texture of a performer's voice supplies an element of character.

From the experiment described in Part 1, the writer found that the blind subjects.

- A. had good understanding of symbols representing movement.
- B. could control the movement of wheels very accurately.
- C. worked together well as a group in solving problems.
- D. got better results than the sighted undergraduates.

the blind not only figured out meanings for each line of motion, but as a group they generally came up with the same meaning at least as frequently as did sighted subjects.

Why does music make us feel? On the one hand, music is a purely abstract art form, devoid of language or explicit ideas. And yet, even though music says little, it still manages to touch us deeply. When listening to our favourite songs, our body betrays all the symptoms of emotional arousal. The pupils in our eyes dilate, our pulse and blood pressure rise, the electrical conductance of our skin is lowered, and the cerebellum, a brain region associated with bodily movement, becomes strangely active. Blood is even re-directed to the muscles in our legs. In other words, sound stirs us at our biological roots.

A recent paper in Nature Neuroscience by a research team in Montreal, Canada, marks an important step in revealing the precise underpinnings of the potent pleasurable stimulus that is music. Although the study involves plenty of fancy technology, including functional magnetic resonance imaging (fMRI) and ligand-based positron emission tomography (PET) scanning, the experiment itself was rather straightforward. After screening 217 individuals who responded to advertisements requesting people who experience 'chills' to instrumental music, the scientists narrowed down the subject pool to ten. They then asked the subjects to bring in their playlist of favourite songs - virtually every genre was represented, from techno to tango - and played them the music while their brain activity was monitored. Because the scientists were combining methodologies (PET and fMRI), they were able to obtain an impressively exact and detailed portrait of music in the brain. The first thing they discovered is that music triggers the production of dopamine - a chemical with a key role in setting people's moods - by the neurons (nerve cells) in both the dorsal and ventral regions of the brain. As these two regions have long been linked with the experience of pleasure, this finding isn't particularly surprising.

What is rather more significant is the finding that the dopamine neurons in the caudate - a region of the brain involved in learning stimulus-response associations, and in anticipating food

and other 'reward' stimuli—were at their most active around 15 seconds before the participants' favourite moments in the music. The researchers call this the 'anticipatory phase' and argue that the purpose of this activity is to help us predict the arrival of our favourite part. The question, of course, is what all these dopamine neurons are up to. Why are they so active in the period preceding the acoustic climax? After all, we typically associate surges of dopamine with pleasure, with the processing of actual rewards. And yet, this cluster of cells is most active when the 'chills' have yet to arrive, when the melodic pattern is still unresolved.

To demonstrate this psychological principle, the musicologist Leonard Meyer, in his classic book *Emotion and Meaning in Music* (1956), analysed the 5th movement of Beethoven's String Quartet in C-sharp minor, Op. 131. Meyer wanted to show how music is defined by its flirtation with—but not submission to—our expectations of order. Meyer dissected 50 measures (bars) of the masterpiece, showing how Beethoven begins with the clear statement of a rhythmic and harmonic pattern and then, in an ingenious tonal dance, carefully holds off repeating it. What Beethoven does instead is suggest variations of the pattern. He wants to preserve an element of uncertainty in his music, making our brains beg for the one chord he refuses to give us. Beethoven saves that chord for the end.

According to Meyer, it is the suspenseful tension of music, arising out of our unfulfilled expectations, that is the source of the music's feeling. While earlier theories of music focused on the way a sound can refer to the real world of images and experiences—its 'connotative' meaning—Meyer argued that the emotions we find in music come from the unfolding events of the music itself. This 'embodied meaning' arises from the patterns the symphony invokes and then ignores. It is this uncertainty that triggers the surge of dopamine in the caudate, as we struggle to figure out what will happen next. We can predict some of the notes, but we can't predict them all, and that is what keeps us listening, waiting expectantly for our reward, for the pattern to be completed.

- 32** What point does the writer emphasise in the first paragraph?
- A how dramatically our reactions to music can vary
 - B how intense our physical responses to music can be
 - C how little we know about the way that music affects us
 - D how much music can tell us about how our brains operate
- 33** What view of the Montreal study does the writer express in the second paragraph?
- A Its aims were innovative.
 - B The approach was too simplistic.
 - C It produced some remarkably precise data.
 - D The technology used was unnecessarily complex.
- 34** What does the writer find interesting about the results of the Montreal study?
- A the timing of participants' neural responses to the music
 - B the impact of the music on participants' emotional state
 - C the section of participants' brains which was activated by the music
 - D the type of music which had the strongest effect on participants' brains
- 35** Why does the writer refer to Meyer's work on music and emotion?
- A to propose an original theory about the subject
 - B to offer support for the findings of the Montreal study
 - C to recommend the need for further research into the subject
 - D to present a view which opposes that of the Montreal researchers
- 36** According to Leonard Meyer, what causes the listener's emotional response to music?
- A the way that the music evokes poignant memories in the listener
 - B the association of certain musical chords with certain feelings
 - C the listener's sympathy with the composer's intentions
 - D the internal structure of the musical composition

三、多选题

- (一) 找出关键信息
- (二) 找定位 (集中出题、分散出题)
- (三) 近义词

C In North America, green roofs have become mainstream, with a wide array of expansive, accessible and food-producing roofs installed in buildings. Again, city leaders and authorities have helped push the movement forward - only recently, San Francisco, USA, created a policy requiring new buildings to have green roofs. Toronto, Canada, has policies dating from the 1990s, encouraging the development of urban farms on rooftops. These countries also benefit from having newer buildings than in many parts of the world, which makes it easier to install green roofs. Being able to keep enough water at roof height and distribute it right across the rooftop is crucial to maintaining the plants on any green roof - especially on 'edible roofs' where fruit and vegetables are farmed. And it's much easier to do this in newer buildings, which can typically hold, greater weight, than to retro-fit old ones. Having a stronger roof also makes it easier to grow a greater variety of plants, since the soil can be.

10–11. Which TWO advantages of using newer buildings for green roofs are mentioned in Paragraph C of the passage?

- A | a longer growing season for edible produce
- B | more economical use of water
- C | greater water-storage capacity
- D | ability to cultivate more plant types
- E | a large surface area for growing plants

四、技巧总结

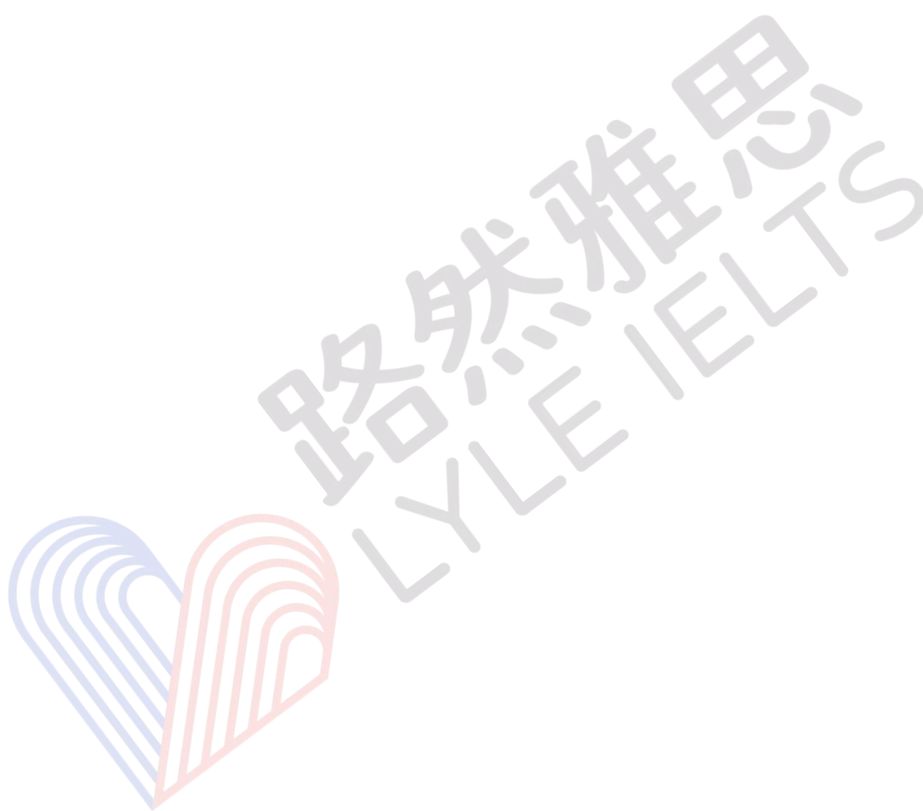
(一) 正确题目的特征

正确选项以改写的形式来出现。

(二) 错误题目的特征

1. 要不就是判断题当中的 F
2. 要不就是判断题之中的 NG

小 Tips: 选项和原文长得越像实际上越有可能是错误答案。



第六章 模拟训练

第一节 模拟训练一

PASSAGE 1

Crop-growing skyscrapers

By the year 2050,nearly 80% of the Earth's population will live in urban centres. Applying the most conservative estimates to current demographic trends, the human population will increase by about three billion people by then. An estimated 10 hectares of new land(about 20% larger than Brazil) will be needed to grow enough food to feed them, if traditional farming methods continue as they are practised today. At present,throughout the world,over80%of the land that is suitable for raising crops is in use.Historically,some 15% of that has been laid waste by poor management practices.What can be done to ensure enough food for the world's population to live on?

The concept of indoor farming is not new, since hothouse production of tomatoes and other produce has been in vogue for sometime.What is new is the urgent need to scale up this technology to accommodate another three billion people. Many believe an entirely new approach to indoor farming is required, employing cutting-edge technologies.One such proposal is for the 'Vertical Farm'.The concept is of multi-storey buildings in which food crops are grown in environmentally controlled conditions.Situated in the heart of urban centres.they would drastically reduce the amount of transportation required to bring food to consumers Vertical farms would need to be efficient, cheap to construct and safe to operate.If successfully implemented,proponents claim, vertical farms offer the promise of urban renewal, sustainable production of a safe and varied food supply through year-round production of all crops),and the eventual repair of ecosystems that have been sacrificed for horizontal farming.

It took humans 10.000 years to learn how to grow most of the crops we now take for granted Along the way we despoiled most of the land we worked, often turning verdant.natural ecozones into semi-arid deserts.Within that same time frame.we evolved into an urban species.in which 60% of the human population now lives vertically in cites.This means that, for the majority,we humans have shelter from the elements, yet we subject our food-bearing plants to the rigours of the great outdoors and can do no more than hope for a good weather year However, more often than not now due to a rapidly changing climate that is not what happens.Massive foods, long droughts.hurricanes and severe monsoons take their toll each year, destroying millions of tons of valuable crops.

The supporters of vertical farming claim many potential advantages for the system.For instance, crops would be produced all year round.as they would be kept in artificially controlled.optimum growing conditions.There would be no weather-related crop failures due to droughts,floods or pests.All the food could be grown organically, eliminating the need for herbicides, pesticides and fertilisers. The system would greatly reduce the incidence of many infectious diseases that are acquired at the agricultural interface. Although the system would consume energy,it would return energy to the grid via methane generation from composting non-edible parts of plants k would also dramatically reduce fossil fuel use.by cutting out the need for tractors, ploughs and shipping.

A major drawback of vertical farming,however, is that the plants would require artificial light.Without it.those plants nearest the windows would be exposed to more sunlight and grow

more quickly, reducing the efficiency of the system. Single-storey greenhouses have the benefit of natural overhead light: even so many still need artificial lighting.

A multi-storey facility with no natural overhead light would require far more. Generating enough light could be prohibitively expensive, unless cheap renewable energy is available, and this appears to be rather a future aspiration than a likelihood for the near future.

One variation on vertical farming that has been developed is to grow plants in stacked trays that move on rails. Moving the trays allows the plants to get enough sunlight. This system is already in operation, and works well within a single-storey greenhouse with light reaching it from above: it is not certain, however, that it can be made to work without that overhead natural light.

Vertical farming is an attempt to address the undoubted problems that we face in producing enough food for a growing population. At the moment, though, more needs to be done to reduce the detrimental impact it would have on the environment particularly as regards the use of energy. While it is possible that much of our food will be grown in skyscrapers in future, most experts currently believe it is far more likely that we will simply use the space available on urban rooftops.

Indoor farming.

1. Some food plants, including _____, are already grown indoors.
2. Vertical farms would be located in _____, meaning that there would be less need to take them long distances to customers.
3. Vertical farms could use methane from plants and animals to produce _____.
4. The consumption of _____ would be cut because agricultural vehicles would be unnecessary.
5. The fact that vertical farms would need _____ light is a disadvantage.
6. One form of vertical farming involves planting in _____ which are not fixed.
7. The most probable development is that food will be grown on _____ in towns and cities.

Questions 8-13

Do the following statements agree with the information given in Reading Passage 1?
in boxes 8-13 on your answer sheet, write

- | | |
|-----------|--|
| TRUE | if the statement agrees with the information |
| FALSE | if the statement contradicts the information |
| NOT GIVEN | if there is no information on this |

8. Methods for predicting the Earth's population have recently changed.
9. Human beings are responsible for some of the destruction to food-producing land.
10. The crops produced in vertical farms will depend on the season.
11. Some damage to food crops is caused by climate change.
12. Fertilisers will be needed for certain crops in vertical farms.
13. Vertical farming will make plants less likely to be affected by infectious diseases.

PASSAGE 2

You should spend about 20 minutes on Questions 14-26, which are based on Reading Passage 2 below

THE FALKIRK WHEEL

A unique engineering achievement

The Falkirk Wheel in Scotland is the world's first and only rotating boat lift. Opened in 2002, it is central to the ambitious £84.5m Millennium Link project to restore navigability across Scotland by reconnecting the historic waterways of the Forth & Clyde and Union Canals.

The major challenge of the project lay in the fact that the Forth & Clyde Canal is situated 35 metres below the level of the Union Canal. Historically, the two canals had been joined near the town of Falkirk by a sequence of 11 locks-enclosed sections of canal in which the water level could be raised or lowered -that stepped down across a distance of 1.5 km. This had been dismantled in 1933, thereby breaking the link. When the project was launched in 1994, the British Waterways authority were keen to create a dramatic twenty-first-century landmark which would not only be a fitting commemoration of the Millennium, but also a lasting symbol of the economic regeneration of the region.

Numerous ideas were submitted for the project, including concepts ranging from rolling eggs to tilting tanks, from giant see-saws to overhead monorails. The eventual winner was a plan for the huge rotating steel boat lift which was to become The Falkirk Wheel. The unique shape of the structure is claimed to have been inspired by various sources, both manmade and natural, most notably a Celtic double-headed axe, but also the vast turning propeller of a ship, the ribcage of a whale or the spine of a fish.

The various parts of The Falkirk Wheel were all constructed and assembled, like one giant toy building set, at Butterley Engineering's Steelworks in Derbyshire, some 400 km from Falkirk. A team there carefully assembled the 1,200 tonnes of steel, painstakingly fitting the pieces together to an accuracy of just 10 mm to ensure a perfect final fit. In the summer of 2001, the structure was then dismantled and transported on 35 lorries to Falkirk, before all being bolted back together again on the ground, and finally lifted into position in five large sections by crane. The Wheel would need to withstand immense and constantly changing stresses as it rotated, so to make the structure more robust, the steel sections were bolted rather than welded together. Over 45,000 bolt holes were matched with their bolts, and each bolt was hand-tightened.

The Wheel consists of two sets of opposing axe-shaped arms, attached about 25 metres apart to a fixed central spine. Two diametrically opposed water-filled 'gondolas', each with a capacity of 360,000 litres, are fitted between the ends of the arms. These gondolas always weigh the same, whether or not they are carrying boats. This is because, according to Archimedes' principle of displacement.

Floating objects displace their own weight in water. So when a boat enters a gondola the amount of water leaving the gondola weighs exactly the same as the boat. This keeps the Wheel balanced and so, despite its enormous mass, it rotates through 180° in five and a half minutes while using very little power. It takes just 1.5 kilowatt-hours (5.4 MJ) of energy to rotate the Wheel - roughly the same as boiling eight small domestic kettles of water.

Boats needing to be lifted up enter the canal basin at the level of the Forth & Clyde Canal and then enter the lower gondola of the Wheel. Two hydraulic steel gates are raised, so as to seal the gondola off from the water in the canal basin. The water between the gates is then pumped out. A hydraulic clamp, which prevents the arms of the Wheel moving while the gondola is docked, is removed, allowing the Wheel to turn. In the central machine room an array of ten hydraulic motors then begins to rotate the central axle. The axle connects to the outer arms of the Wheel, which

begin to rotate at a speed of 1/8 of a revolution per minute. As the wheel rotates, the gondolas are kept in the upright position by a simple gearing system. Two eight-metre-wide cogs orbit a fixed inner cog of the same width, connected by two smaller cogs travelling in the opposite direction to the outer cogs—so ensuring that the gondolas always remain level. When the gondola reaches the top, the boat passes straight onto the aqueduct situated 24 metres above the canal basin.

The remaining 11 metres of lift needed to reach the Union Canal is achieved by means of a pair of locks. The Wheel could not be constructed to elevate boats over the full 35-metre difference between the two canals, owing to the presence of the historically important Antonine Wall, which was built by the Romans in the second century AD. Boats travel under this wall via a tunnel, then through the locks, and finally on to the Union Canal.

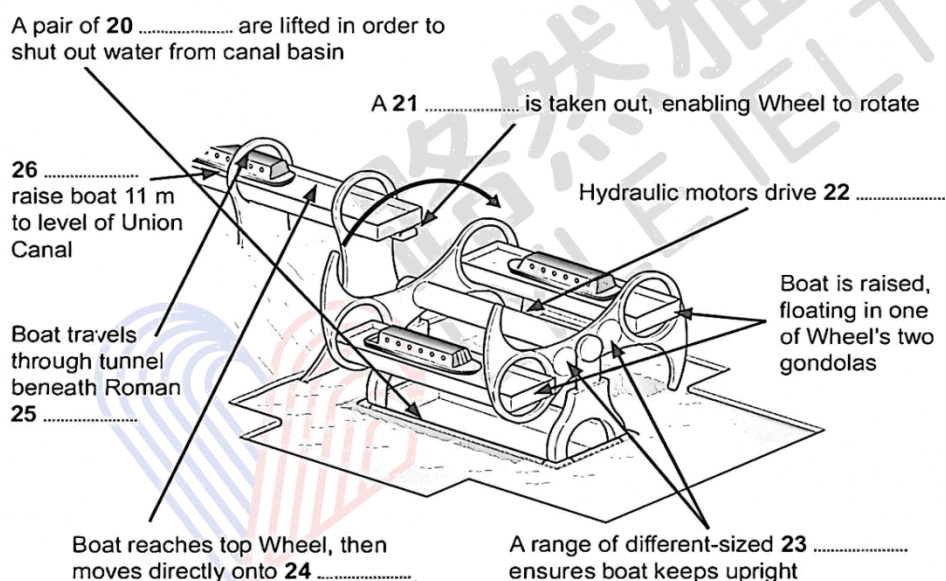
Questions 20–26

Label the diagram below.

Choose **ONE WORD** from the passage for each answer.

Write your answers in boxes 20–26 on your answer sheet.

How a boat is lifted on the Falkirk Wheel



Questions 14–19

Do the following statements agree with the information given in Reading Passage 2?

In boxes 14–19 on your answer sheet, write

TRUE if the statement agrees with the information
FALSE if the statement contradicts the information
NOT GIVEN if there is no information on this

- 14 The Falkirk Wheel has linked the Forth & Clyde Canal with the Union Canal for the first time in their history.

- 15 There was some opposition to the design of the Falkirk Wheel at first.
- 16 The Falkirk Wheel was initially put together at the location where its components were manufactured.
- 17 The Falkirk Wheel is the only boat lift in the world which has steel sections bolted together by hand.
- 18 The weight of the gondolas varies according to the size of boat being carried.
- 19 The construction of the Falkirk Wheel site took into account the presence of a nearby ancient monument.

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40, which are based on Reading Passage 3 below.

Reducing the Effects of Climate Change

Mark Rowe reports on the increasingly ambitious geo-engineering projects being explored by scientists

A Such is our dependence on fossil fuels, and such is the volume of carbon dioxide already released into the atmosphere, that many experts agree that significant global warming is now inevitable. They believe that the best we can do is keep it at a reasonable level, and at present the only serious option for doing this is cutting back on our carbon emissions. But while a few countries are making major strides in this regard, the majority are having great difficulty even stemming the rate of increase, let alone reversing it. Consequently, an increasing number of scientists are beginning to explore the alternative of geo-engineering—a term which generally refers to the intentional large-scale manipulation of the environment. According to its proponents, geo-engineering is the equivalent of a backup generator: if Plan A—reducing our dependency on fossil fuels—fails, we require a Plan B, employing grand schemes to slow down or reverse the process of global warming.

B Geo-engineering has been shown to work, at least on a small localised scale. For decades, May Day parades in Moscow have taken place under clear blue skies, aircraft having deposited dry ice, silver iodide and cement powder to disperse clouds. Many of the schemes now suggested look to do the opposite, and reduce the amount of sunlight reaching the planet. The most eye-catching idea of all is suggested by Professor Roger Angel of the University of Arizona. His scheme would employ up to 16 trillion minute spacecraft, each weighing about one gram, to form a transparent, sunlight-refracting sunshade in an orbit 1.5 million km above the Earth. This could, argues Angel, reduce the amount of light reaching the Earth by two per cent.

C The majority of geo-engineering projects so far carried out—which include planting forests in deserts and depositing iron in the ocean to stimulate the growth of algae—have focused on achieving a general cooling of the Earth. But some look specifically at reversing the melting at the poles, particularly the Arctic. The reasoning is that if you replenish the ice sheets and frozen waters of the high latitudes, more light will be reflected back into space, so reducing the warming of the oceans and atmosphere.

D The concept of releasing aerosol sprays into the stratosphere above the Arctic has been proposed by several scientists. This would involve using sulphur or hydrogen sulphide aerosols so that sulphur dioxide would form clouds, which would, in turn, lead to a global dimming. The idea is modelled on historic volcanic explosions, such as that of Mount Pinatubo in the Philippines in

1991, which led to a short-term cooling of global temperatures by 0.5°C. Scientists have also scrutinised whether it's possible to preserve the ice sheets of Greenland with reinforced high-tension cables, preventing icebergs from moving into the sea. Meanwhile in the Russian Arctic, geo-engineering plans include the planting of millions of birch trees. Whereas the region's native evergreen pines shade the snow and absorb radiation, birches would shed their leaves in winter, thus enabling radiation to be reflected by the snow. Re-routing Russian rivers to increase cold water flow to ice-forming areas could also be used to slow down warming, say some climate scientists.

E But will such schemes ever be implemented? Generally speaking, those who are most cautious about geo-engineering are the scientists involved in the research. Angel says that his plan is 'no substitute for developing renewable energy: the only permanent solution'. And Dr Phil Rasch of the US-based Pacific Northwest National Laboratory is equally guarded about the role of geo-engineering: 'I think all of us agree that if we were to end geo-engineering on a given day, then the planet would return to its pre-engineered condition very rapidly, and probably within ten to twenty years. That's certainly something to worry about.'

F The US National Center for Atmospheric Research has already suggested that the proposal to inject sulphur into the atmosphere might affect rainfall patterns across the tropics and the Southern Ocean. 'Geo-engineering plans to inject stratospheric aerosols or to seed clouds would act to cool the planet, and act to increase the extent of sea ice,' says Rasch. 'But all the models suggest some impact on the distribution of precipitation.'

G A further risk with geo-engineering projects is that you can "overshoot", 'says Dr Dan Lunt, from the University of Bristol's School of Geophysical Sciences, who has studied the likely impacts of the sunshade and aerosol schemes on the climate. 'You may bring global temperatures back to pre-industrial levels, but the risk is that the poles will still be warmer than they should be and the tropics will be cooler than before industrialisation.' To avoid such a scenario, Lunt says Angel's project would have to operate at half strength; all of which reinforces his view that the best option is to avoid the need for geo-engineering altogether.'

H The main reason why geo-engineering is supported by many in the scientific community is that most researchers have little faith in the ability of politicians to agree and then bring in the necessary carbon cuts. Even leading conservation organisations see the value of investigating the potential of geo-engineering. According to Dr Martin Sommerkorn, climate change advisor for the World Wildlife Fund's International Arctic Programme, 'Human-induced climate change has brought humanity to a position where we shouldn't exclude thinking thoroughly about this topic and its possibilities.'

Questions 27–29

Reading Passage 3 has eight paragraphs **A–H**.

Which paragraph contains the following information?

Write the correct letter, **A–H**, in boxes 27–29 on your answer sheet.

- 27 mention of a geo-engineering project based on an earlier natural phenomenon
- 28 an example of a successful use of geo-engineering
- 29 a common definition of geo-engineering

Questions 30–36

Complete the table below.

Choose **ONE WORD** from the passage for each answer.

Write your answers in boxes 30–36 on your answer sheet.

GEO-ENGINEERING PROJECTS

Procedure	Aim
put a large number of tiny spacecraft into orbit far above Earth	to create a 30 that would reduce the amount of light reaching Earth
place 31 in the sea	to encourage 32 to form
release aerosol sprays into the stratosphere	to create 33 that would reduce the amount of light reaching Earth
fix strong 34 to Greenland ice sheets	to prevent icebergs moving into the sea
plant trees in Russian Arctic that would lose their leaves in winter	to allow the 35 to reflect radiation
change the direction of 36	to bring more cold water into ice-forming areas

Questions 37–40

Look at the following statements (Questions 37–40) and the list of scientists below.

Match each statement with the correct scientist, **A–D**.

Write the correct letter, **A–D**, in boxes 37–40 on your answer sheet.

- 37 The effects of geo-engineering may not be long-lasting.
- 38 Geo-engineering is a topic worth exploring.
- 39 It may be necessary to limit the effectiveness of geo-engineering projects.
- 40 Research into non-fossil-based fuels cannot be replaced by geo-engineering.

List of Scientists

- A** Roger Angel
- B** Phil Rasch
- C** Dan Lunt
- D** Martin Sommerkorn

第二节 模拟训练二

READING READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13, which are based on Reading Passage 1 below

The History of Glass

From our earliest origins, man has been making use of glass. Historians have discovered that a type of natural glass-obsidian -formed in places such as the mouth of a volcano as a result of the intense heat of an eruption melting sand -was first used as tips for spears. Archaeologists have even found evidence of man-made glass which dates back to 4000 BC; this took the form of glazes used for coating stone beads. It was not until 1500 BC, however, that the first hollow glass container was made by covering a sand core with a layer of molten glass.

Glass blowing became the most common way to make glass containers from the first century BC. The glass made during this time was highly coloured due to the impurities of the raw material. In the first century AD, methods of creating colourless glass were developed, which was then tinted by the addition of colouring materials. The secret of glass making was taken across Europe by the Romans during this century. However, they guarded the skills and technology required to make glass very closely, and it was not until their empire collapsed in 476 AD that glass-making knowledge became widespread throughout Europe and the Middle East. From the 10th century onwards, the Venetians gained a reputation for technical skill and artistic ability in the making of glass bottles, and many of the city's craftsmen left Italy to set up glass works throughout Europe.

A major milestone in the history of glass occurred with the invention of lead crystal glass by the English glass manufacturer George Ravenscroft (1632-1683). He attempted to counter the effect of clouding that sometimes occurred in blown glass by introducing lead to the raw materials used in the process. The new glass he created was softer and easier to decorate, and had a higher refractive index, adding to its brilliance and beauty, and it proved invaluable to the optical industry. It is thanks to Ravenscroft's invention that optical lenses, astronomical telescopes, microscopes and the like became possible.

In Britain, the modern glass industry only really started to develop after the repeal of the Excise Act in 1845. Before that time, heavy taxes had been placed on the amount of glass melted in a glasshouse, and were levied continuously from 1745 to 1845. Joseph Paxton's Crystal Palace at London's Great Exhibition of 1851 marked the beginning of glass as a material used in the building industry. This revolutionary new building encouraged the use of glass in public, domestic and horticultural architecture. Glass manufacturing techniques also improved with the advancement of science and the development of better technology.

From 1887 onwards, glass making developed from traditional mouth-blowing to a semi-automatic process, after factory-owner HM Ashley introduced a machine capable of producing 200 bottles per hour in Castleford, Yorkshire, England - more than three times quicker than any previous production method. Then in 1907, the first fully automated machine was developed in the USA by Michael Owens - founder of the Owens Bottle Machine Company (later the major manufacturers Owens-Illinois) - and installed in its factory. Owens' invention could produce an impressive 2,500 bottles per hour. Other developments followed rapidly, but it was not until the First World War, when Britain became cut off from essential glass suppliers, that glass

became part of the scientific sector. Previous to this, glass had been seen as a craft rather than a precise science.

Today, glass making is big business. It has become a modern, hi-tech industry operating in a fiercely competitive global market where quality, design and service levels are critical to maintaining market share. Modern glass plants are capable of making millions of glass containers a day in many different colours, with green, brown and clear remaining the most popular. Few of us can imagine modern life without glass. It features in almost every aspect of our lives - in our homes, our cars and whenever we sit down to eat or drink. Glass packaging is used for many products, many beverages are sold in glass, as are numerous foodstuffs, as well as medicines and cosmetics.

Glass is an ideal material for recycling, and with growing consumer concern for green issues, glass bottles and jars are becoming ever more popular. Glass recycling is good news for the environment. It saves used glass containers being sent to land fill. As less energy is needed to melt recycled glass than to melt down raw materials, this also saves fuel and production costs. Recycling also reduces the need for raw materials to be quarried, thus saving precious resources.

Questions 1–8

Complete the notes below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 1–8 on your answer sheet.

The History of Glass

- Early humans used a material called **1** to make the sharp points of their **2**
- 4000 BC: **3** made of stone were covered in a coating of man-made glass.
- First century BC: glass was coloured because of the **4** in the material.
- Until 476 AD: Only the **5** knew how to make glass.
- From 10th century: Venetians became famous for making bottles out of glass.
- 17th century: George Ravenscroft developed a process using **6** to avoid the occurrence of **7** in blown glass.
- Mid-19th century: British glass production developed after changes to laws concerning **8**

Questions 9–13

In boxes 9–13 on your answer sheet, write

TRUE if the statement agrees with the information
FALSE if the statement contradicts the information
NOT GIVEN if there is no information on this

- 9 In 1887, HM Ashley had the fastest bottle-producing machine that existed at the time.
- 10 Michael Owens was hired by a large US company to design a fully-automated bottle manufacturing machine for them.
- 11 Nowadays, most glass is produced by large international manufacturers.
- 12 Concern for the environment is leading to an increased demand for glass containers.
- 13 It is more expensive to produce recycled glass than to manufacture new glass.

第三节 模拟训练三

You should spend about 20 minutes on Questions 14–26, which are based on Reading Passage 2 below.

Bring back the big cats

It's time to start returning vanished native animals to Britain, says John Vesty

There is a poem, written around 598 AD, which describes hunting a mystery animal called a llewyn. But what was it? Nothing seemed to fit, until 2006, when an animal bone, dating from around the same period, was found in the Kinsey Cave in northern England. Until this discovery, the lynx—a large spotted cat with tasselled ears—was presumed to have died out in Britain at least 6,000 years ago, before the inhabitants of these islands took up farming. But the 2006 find, together with three others in Yorkshire and Scotland, is compelling evidence that the lynx and the mysterious llewyn were in fact one and the same animal. If this is so, it would bring forward the tassel-eared cat's estimated extinction date by roughly 5,000 years.

However, this is not quite the last glimpse of the animal in British culture. A 9th-century stone cross from the Isle of Eigg shows, alongside the deer, boar and aurochs pursued by a mounted hunter, a speckled cat with tasselled ears. Were it not for the animal's backside having worn away with time, we could have been certain, as the lynx's stubby tail is unmistakable. But even without this key feature, it's hard to see what else the creature could have been. The lynx is now becoming the totemic animal of a movement that is transforming British environmentalism: rewilding.

Rewilding means the mass restoration of damaged ecosystems. It involves letting trees return to places that have been denuded, allowing parts of the seabed to recover from trawling and dredging, permitting rivers to flow freely again. Above all, it means bringing back missing species. One of the most striking findings of modern ecology is that ecosystems without large predators behave in completely different ways from those that retain them. Some of them drive dynamic processes that resonate through the whole food chain, creating niches for hundreds of species that might otherwise struggle to survive. The killers turn out to be bringers of life.

Such findings present a big challenge to British conservation, which has often selected arbitrary assemblages of plants and animals and sought, at great effort and expense, to prevent them from changing. It has tried to preserve the living world as if it were a jar of pickles, letting nothing in and nothing out, keeping nature in a state of arrested development. But ecosystems are not merely collections of species; they are also the dynamic and ever-shifting relationships between them. And this dynamism often depends on large predators.

At sea the potential is even greater: by protecting large areas from commercial fishing, we could once more see what 18th-century literature describes: vast shoals of fish being chased by fin and sperm whales, within sight of the English shore. This policy would also greatly boost catches in the surrounding seas; the fishing industry's insistence on scouring every inch of seabed, leaving no breeding reserves, could not be more damaging to its own interests.

Rewilding is a rare example of an environmental movement in which campaigners articulate what they are for rather than only what they are against. One of the reasons why the enthusiasm for rewilding is spreading so quickly in Britain is that it helps to create a more inspiring vision than the green movement's usual promise of 'Follow us and the world will be slightly less awful than it would otherwise have been.'

The lynx presents no threat to human beings: there is no known instance of one preying on people. It is a specialist predator of roe deer, a species that has exploded in Britain in recent decades, holding back, by intensive browsing, attempts to re-establish forests. It will also wrinkle out sika deer: an exotic species that is almost impossible for human beings to control, as it hides in impenetrable plantations of young trees. The attempt to reintroduce this predator marries well with the aim of bringing forests back to parts of our bare and barren uplands. The lynx requires deep cover, and as such presents little risk to sheep and other livestock, which are supposed, as a condition of farm subsidies, to be kept out of the woods.

On a recent trip to the Cairngorm Mountains, I heard several conservationists suggest that the lynx could be reintroduced there within 20 years. If trees return to the bare hills elsewhere in Britain, the big cats could soon follow. There is nothing extraordinary about these proposals, seen from the perspective of any where else in Europe. The lynx has now been reintroduced to the Jura Mountains, the Alps, the Vosges in eastern France and the Harz mountains in Germany, and has re-established itself in many more places. The European population has tripled since 1970 to roughly 10,000. As with wolves, bears, beavers, boar, bison, moose and many other species, the lynx has been able to spread as farming has left the hills and people discover that it is more lucrative to protect charismatic wildlife than to hunt it, as tourists will pay for the chance to see it. Large-scale rewilding is happening almost everywhere - except Britain.

Here, attitudes are just beginning to change. Conservationists are starting to accept that the old preservation-jar model is failing, even on its own terms. Already, projects such as Trees for Life in the Highlands provide a hint of what might be coming. An organisation is being set up that will seek to catalyse the rewilding of land and sea across Britain, its aim being to reintroduce that rarest of species to British ecosystems: hope.

Questions 14–18

Write the correct letter, **A**, **B**, **C** or **D**, in boxes 14–18 on your answer sheet.

- 14 What did the 2006 discovery of the animal bone reveal about the lynx?
- A Its physical appearance was very distinctive.
 - B Its extinction was linked to the spread of farming.
 - C It vanished from Britain several thousand years ago.
 - D It survived in Britain longer than was previously thought.
- 15 What point does the writer make about large predators in the third paragraph?
- A Their presence can increase biodiversity.
 - B They may cause damage to local ecosystems.
 - C Their behaviour can alter according to the environment.
 - D They should be reintroduced only to areas where they were native.
- 16 What does the writer suggest about British conservation in the fourth paragraph?
- A It has failed to achieve its aims.
 - B It is beginning to change direction.
 - C It has taken a misguided approach.
 - D It has focused on the most widespread species.
- 17 Protecting large areas of the sea from commercial fishing would result in
- A practical benefits for the fishing industry.
 - B some short-term losses to the fishing industry.
 - C widespread opposition from the fishing industry.
 - D certain changes to techniques within the fishing industry.
- 18 According to the author, what distinguishes rewilding from other environmental campaigns?
- A Its objective is more achievable.
 - B Its supporters are more articulate.
 - C Its positive message is more appealing.
 - D It is based on sounder scientific principles.

Questions 19–22

Complete the summary using the list of words and phrases **A–F** below.

Write the correct letter, **A–F**, in boxes 19–22 on your answer sheet.

Reintroducing the lynx to Britain

There would be many advantages to reintroducing the lynx to Britain. While there is no evidence that the lynx has ever put **19** in danger, it would reduce the numbers of certain **20** whose populations have increased enormously in recent decades. It would present only a minimal threat to **21** , provided these were kept away from lynx habitats. Furthermore, the reintroduction programme would also link efficiently with initiatives to return native **22** to certain areas of the country.

- | | | |
|-----------------------|-----------------------------|-----------------------|
| A trees | B endangered species | C hillsides |
| D wild animals | E humans | F farm animals |

Questions 23–26

Do the following statements agree with the claims of the writer in Reading Passage 2?

In boxes 23–26 on your answer sheet, write

- YES** if the statement agrees with the claims of the writer
NO if the statement contradicts the claims of the writer
NOT GIVEN if it is impossible to say what the writer thinks about this

- 23** Britain could become the first European country to reintroduce the lynx.
24 The large growth in the European lynx population since 1970 has exceeded conservationists' expectations.
25 Changes in agricultural practices have extended the habitat of the lynx in Europe.
26 It has become apparent that species reintroduction has commercial advantages.

READING PASSAGE 3

You should spend about 20 minutes on **Questions 27–40**, which are based on Reading Passage 3 on pages 89 and 90.

Questions 27–33

Reading Passage 3 has seven paragraphs, **A–G**.

Choose the correct heading for each paragraph from the list of headings below.

Write the correct number, **i–viii**, in boxes 27–33 on your answer sheet.

List of Headings

- i Disputes over financial arrangements regarding senior managers
- ii The impact on companies of being subjected to close examination
- iii The possible need for fundamental change in every area of business
- iv Many external bodies being held responsible for problems
- v The falling number of board members with broad enough experience
- vi A risk that not all directors take part in solving major problems
- vii Boards not looking far enough ahead
- viii A proposal to change the way the board operates

27 Paragraph A

28 Paragraph B

29 Paragraph C

30 Paragraph D

31 Paragraph E

32 Paragraph F

33 Paragraph G

UK companies need more effective boards of directors

A After a number of serious failures of governance (that is, how they are managed at the highest level), companies in Britain, as well as elsewhere, should consider radical changes to their directors' roles. It is clear that the role of a board director today is not an easy one. Following the 2008 financial meltdown, which resulted in a deeper and more prolonged period of economic downturn than anyone expected, the search for explanations in the many post-mortems of the crisis has meant blame has been spread far and wide. Governments, regulators, central banks and auditors have all been in the frame. The role of bank directors and management and their widely publicised failures have been extensively picked over and examined in reports, inquiries and commentaries.

B The knock-on effect of this scrutiny has been to make the governance of companies in general an issue of intense public debate and has significantly increased the pressures on, and the responsibilities of, directors. At the simplest and most practical level, the time involved in fulfilling the demands of a board directorship has increased significantly, calling into question the effectiveness of the classic model of corporate governance by part-time, independent non-executive directors. Where once a board schedule may have consisted of between eight and ten meetings a year, in many companies the number of events requiring board input and decisions has dramatically risen. Furthermore, the amount of reading and preparation required for each meeting is increasing. Agendas can become overloaded and this can mean the time for constructive debate must necessarily be restricted in favour of getting through the business.

C Often, board business is devolved to committees in order to cope with the

workload, which may be more efficient but can mean that the board as a whole is less involved in fully addressing some of the most important issues. It is not uncommon for the audit committee meeting to last longer than the main board meeting itself. Process may take the place of discussion and be at the expense of real collaboration, so that boxes are ticked rather than issues tackled.

D A radical solution, which may work for some very large companies whose businesses are extensive and complex, is the professional board, whose members would work up to three or four days a week, supported by their own dedicated staff and advisers. There are obvious risks to this and it would be important to establish clear guidelines for such a board to ensure that it did not step on the toes of management by becoming too engaged in the day-to-day running of the company. Problems of recruitment, remuneration and independence could also arise and this structure would not be appropriate for all companies. However, more professional and better-informed boards would have been particularly appropriate for banks where the executives had access to information that part-time non-executive directors lacked, leaving the latter unable to comprehend or anticipate the 2008 crash.

E One of the main criticisms of boards and their directors is that they do not focus sufficiently on longer-term matters of strategy, sustainability and governance, but instead concentrate too much on short-term financial metrics. Regulatory requirements and the structure of the market encourage this behaviour. The tyranny of quarterly reporting can distort board decision-making, as directors have to 'make the numbers' every four months to meet the insatiable appetite of the market for more data. This serves to encourage the trading methodology of a certain kind of investor who moves in and out of a stock without engaging in constructive dialogue with the company about strategy or performance, and is simply seeking a short-term financial gain. This effect has been made worse by the changing profile of investors due to the globalisation of capital and the increasing use of automated trading systems. Corporate culture adapts and management teams are largely incentivised to meet financial goals.

F Compensation for chief executives has become a combat zone where pitched battles between investors, management and board members are fought, often behind closed doors but increasingly frequently in the full glare of press attention. Many would argue that this is in the interest of transparency and good governance as shareholders use their muscle in the area of pay to pressure boards to remove under performing chief executives. Their powers to vote down executive remuneration policies increased when binding votes came into force. The chair of the remuneration committee can be an exposed and lonely role, as Alison Carnwath, chair of Barclays Bank's remuneration committee, found when she had to resign, having been roundly criticised for trying to defend the enormous bonus to be paid to the chief executive; the irony being that she was widely understood to have spoken out against it in the privacy of the committee.

G The financial crisis stimulated a debate about the role and purpose of the company and a heightened awareness of corporate ethics. Trust in the corporation has been eroded and academics such as Michael Sandel, in his thoughtful and bestselling book *What Money Can't Buy*, are questioning the morality of capitalism and the market economy. Boards of companies in all sectors will need to widen their perspective to encompass these issues and this may involve a realignment of corporate goals. We live in challenging times.

Questions 34–37

Do the following statements agree with the claims of the writer in Reading Passage 3?

In boxes 34–37 on your answer sheet, write

- YES** if the statement agrees with the claims of the writer
NO if the statement contradicts the claims of the writer
NOT GIVEN if it is impossible to say what the writer thinks about this

- 34** Close scrutiny of the behaviour of boards has increased since the economic downturn.
- 35** Banks have been mismanaged to a greater extent than other businesses.
- 36** Board meetings normally continue for as long as necessary to debate matters in full.
- 37** Using a committee structure would ensure that board members are fully informed about significant issues.

Questions 38–40

Complete the sentences below.

Choose **ONE WORD ONLY** from the passage for each answer.

Write your answers in boxes 38–40 on your answer sheet.

- 38** Before 2008, non-executive directors were at a disadvantage because of their lack of
- 39** Boards tend to place too much emphasis on considerations that are only of short-term relevance.
- 40** On certain matters, such as pay, the board may have to accept the views of